

Leen's PhD Defense



Hi Leen! We have arrived in the Justus Lipsius room!



KU LEUVEN

WORDS DIVIDE PICTOGRAPHS UNITE

Pictograph Communication Technologies
for People with an Intellectual Disability

Leen Sevens

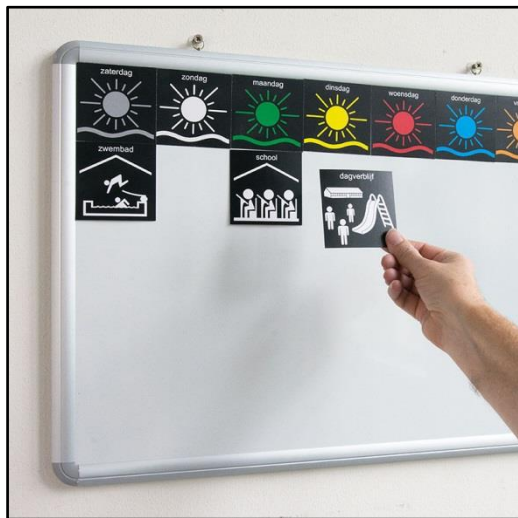
Supervisor: Prof. Dr. Frank Van Eynde

Co-supervisor: Dr. Vincent Vandeghinste





Introduction

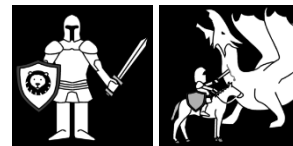




“De ridder doodde de draak”

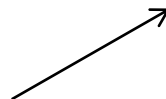
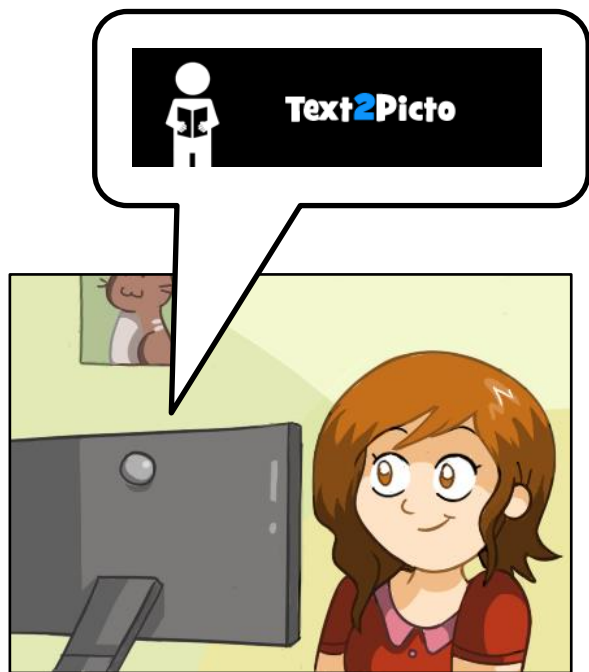


Sclera

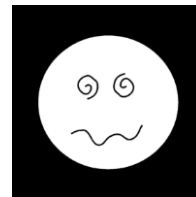


Beta

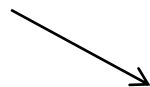




Spelling



Syntax



Semantics

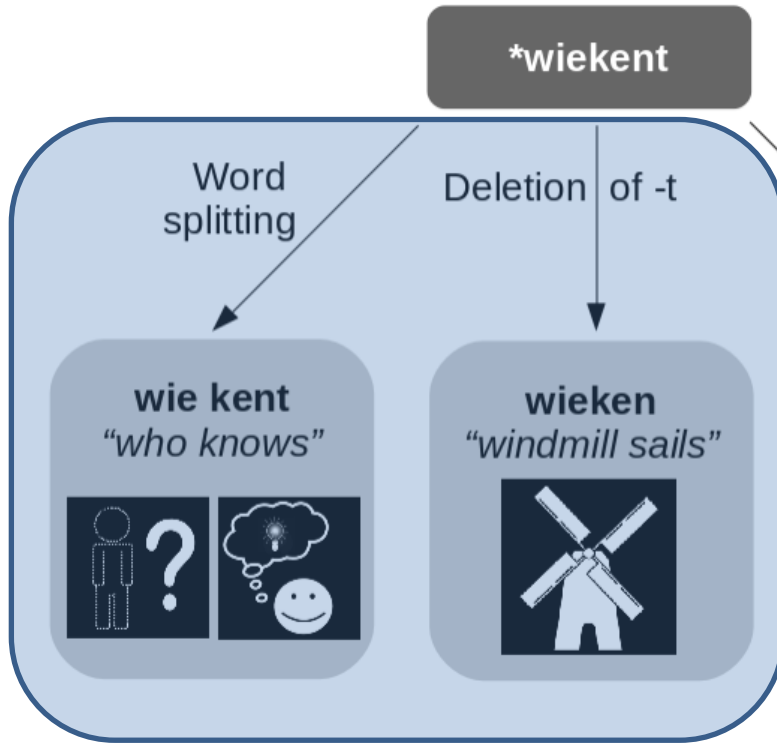


Input: volgend wiekent ga ik naarhuis

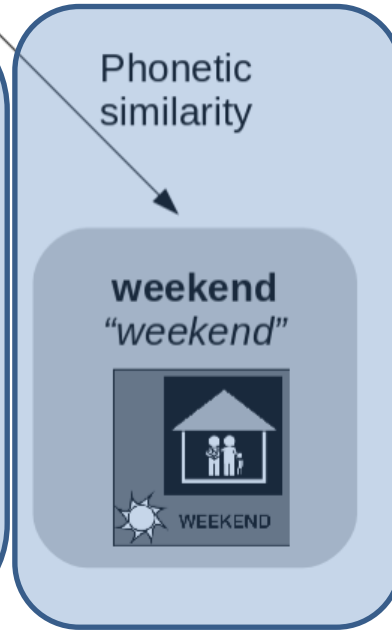
Correction by original Text2Picto: volgend wieken ga ik naarhuis



Typographic errors



Cognitive errors

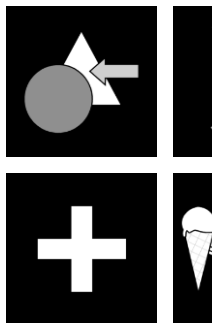




Text2Picto

Improvement #2: Syntactic simplification

Na de verdediging ga
'After the defense we



dit gebouw.
lobby of this building.'



De post
'The ma



De postbode, die gisteren door onze hond gebeten werd,
The mailman, who yesterday by our dog bitten was,

eist dat we een nieuwe broek voor hem kopen.
demands that we a new pair-of-pants for him buy.

"The mailman, who was bitten by our dog last week, demands that we buy him a new pair of pants."

De postbode eist
The mailman demands



(b)

die gisteren door onze hond gebeten werd
who yesterday by our dog bitten was
(Antecedent: postbode "mailman", SUBJ)

(f)

(h) - (i) - (j)

postbode gebeten werd door onze hond gisteren
mailman bitten was by our dog yesterday

dat we een nieuwe broek voor hem kopen
that we a new pair-of-pants for him buy

(i)

(dat) we kopen een nieuwe broek voor hem
(that) we buy a new pair-of-pants for him



(k)

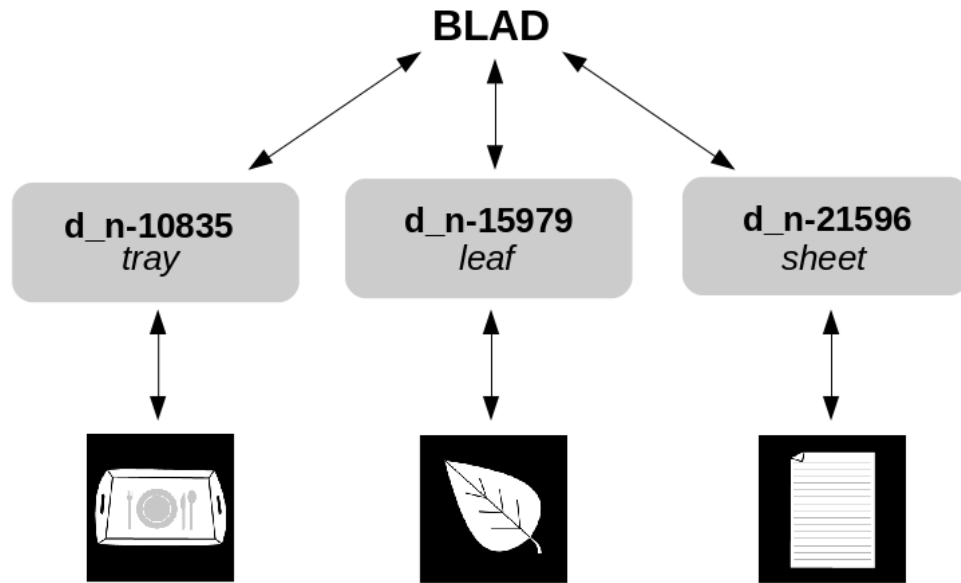
hond gebeten postbode gisteren
dog bitten mailman yesterday





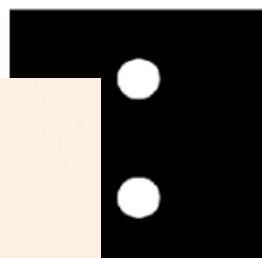
Text2Picto

Improvement #3: Word sense disambiguation





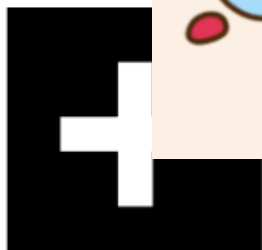
een rec
a recip



...



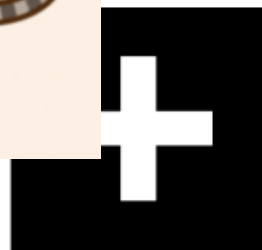
bloem
flower



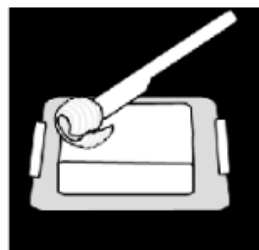
en
and



eieren
eggs



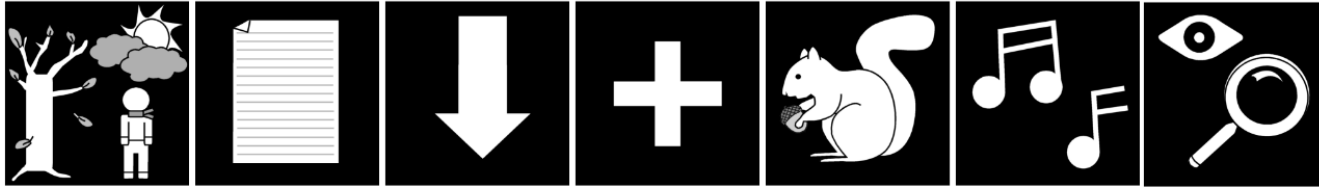
en
and



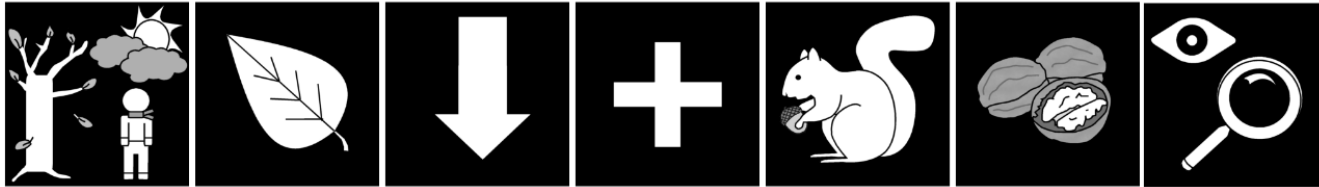
boter
butter



Most frequent sense



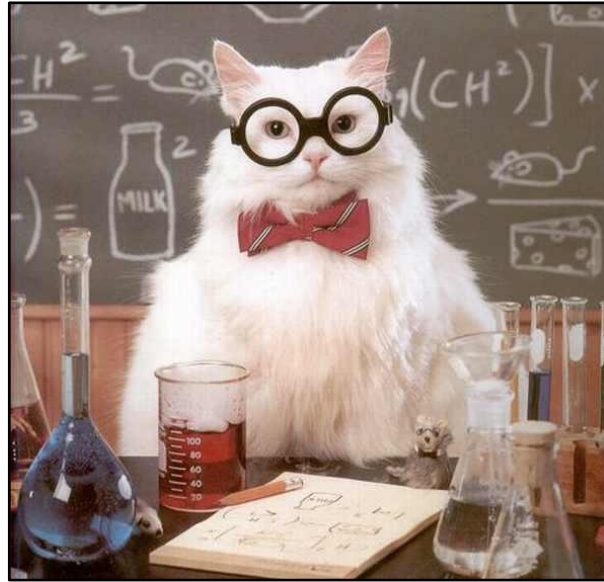
WSD tool



(...) dat in de herfst de blaadjes vallen en de eekhoorn nootjes zoekt.

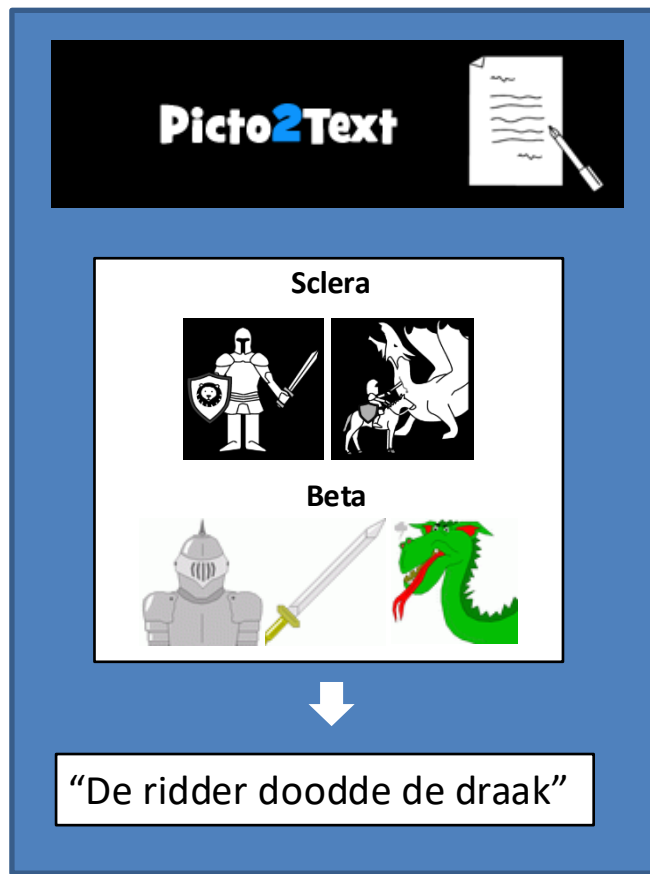
'(...) that, in the autumn, the leaves fall down and the squirrel looks for nuts.'





Evaluations



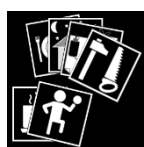


Pictograph interface



Translation into text

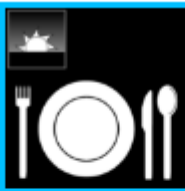
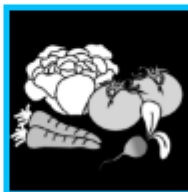




Picto2Text

Development of a pictograph interface







Predictions

The user inserts
pictographs



The prediction tool suggests
new pictographs



to bark, to eat, to be,...



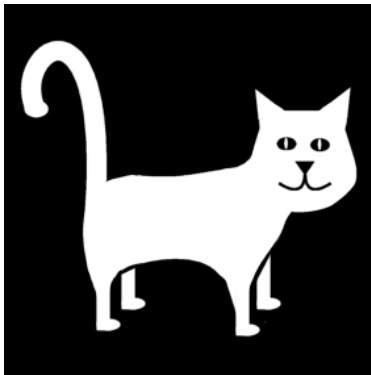


Picto2Text

Translation into text

Challenges

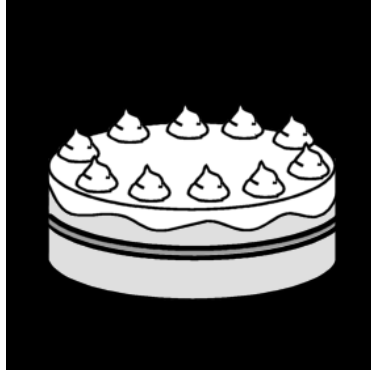
A pictograph can correspond to different (synonymous) words



Kat, poes, huistijger,...



Most pictographs make no distinction between singular and plural



Taart, taartje, taarten,...



Verbs are not inflected, and there are no temporal auxiliaries



dans, dansen, danste, dansten, dansend,...





5-gram LM huge: Hallo je prachtige honden hoe naam hond ben Lassie?

5-gram LM CGN: Hallo je mooie hond hoe naam hond is Lassie?

5-gram LM SONAR: Hallo jij Mooi honden Hoe naam hond is lassie?

LSTM CGN: Hallo jij mooi de honden hoe de naam de honden zijn Lassie?

Moses Broken: Hallo zoals jij hebben een mooie hond, hoe het op naam van de hond is een Lassie?

Moses T2P: Hallo, een mooie hond, hoe heet het hondje is een Lassie?

OpenNMT Demolish: Hallo, jij hebben mooie hond, hoe de naam van de hond is, Lassie?

OpenNMT T2P (BRNN): Hallo, je hebt een mooi hondje hoe heet de hond is dat een Lassie?



Curious?



<http://picto.ccl.kuleuven.be/>



Thank you!

